

Line Following Robot for Lab Material Transport

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Abstract—The transportation of laboratory materials within educational and research facilities presents significant challenges in terms of efficiency, safety, and resource allocation. This paper presents the design, implementation, and evaluation of line following robot specifically developed for laboratory material transport applications. The system includes an Arduino microcontroller with infrared (IR) sensors for line detection and ultrasonic sensors for obstacle avoidance, creating a cost-effective and reliable autonomous navigation solution. The robot follows predefined paths marked by black lines on the floor while avoiding obstacles in its path. This system demonstrates the feasibility of this solution in reducing manual material handling labour while maintaining material safety and delivery accuracy. Effectiveness of the system is validated through experimental trials in a simulated laboratory environment, with results showing high successful delivery rate and autonomous operation capability. This system contributes to the broader application of robotics in operational efficiency within academic institutions and provides a scalable framework for institutions seeking automation solutions.

Index Terms—Autonomous robot, line following, Arduino, IR sensors, ultrasonic sensor, laboratory automation, material transport, obstacle avoidance

I. INTRODUCTION

Laboratory environments in educational institutions and research facilities require frequent transportation of materials, equipment, chemicals, and samples between different locations. This task is traditionally performed manually by laboratory personnel, which have several challenges. The reliance on manual transport represents a significant misallocation of human resources, as highly trained laboratory technicians and scientists must spend valuable time on routine material delivery tasks instead of focusing on their core expertise. Additionally, the operational costs associated with manual material handling including labor expenses, potential errors requiring rework, and inefficient workflow scheduling create a significant economic burden on laboratory operations. The increasing demand for laboratory automation has driven the development of autonomous mobile robots capable of performing material transport tasks with minimal amount of human intervention.

Line following robots are fundamental class of autonomous mobile robots that navigate by tracking black lines marked on the floor surface. These robots have found applications in various domains including warehouse automation, manufacturing assembly lines, and hospital material transport systems [1].

The integration of obstacle avoidance capabilities further enhances the practical applicability of such systems in dynamic environments where unexpected obstacles may appear in the robot's path.

This paper presents the development of an Arduino-based line following robot specifically designed for laboratory material transport. The system employs infrared sensors for line detection and ultrasonic sensors for obstacle avoidance, providing a comprehensive autonomous navigation solution. The key contributions of this work include:

- Design of a low-cost, robust line following robot using readily available components
- Implementation of a hybrid navigation algorithm combining line tracking and obstacle avoidance
- Experimental validation in a laboratory environment with performance analysis
- Evaluation of system reliability and energy efficiency for continuous operation

II. LITERATURE REVIEW

A. Line Following Robot System

Line following robots have been extensively researched over the past decades, with various sensing technologies and control methodologies proposed. Pakdaman and Sanaatiyan [2] presented one of the microcontroller-based designs for line detection. More recent implementations have utilized infrared sensor arrays to achieve higher accuracy and faster response times [3].

The application of PID (Proportional-Integral-Derivative) control for line following has been demonstrated to significantly improve tracking performance compared to simple on-off control strategies [4].

B. Arduino-Based Mobile Robotics

Arduino microcontrollers have become popular platforms for educational and prototype robotic systems due to their ease of programming, extensive community support, and cost-effectiveness.

Mahmud et al. presented the design and implementation of an ultrasonic sensor-based obstacle avoidance system for Arduino-based mobile robots. The proposed system utilizes ultrasonic distance measurements to detect obstacles in real time and dynamically adjust the robot's motion to avoid collisions. The study demonstrates a low-cost and efficient

approach suitable for indoor robotic navigation, emphasizing simplicity in hardware design and reliability in obstacle detection. Experimental results confirm that the system effectively improves autonomous navigation performance in structured environments, making it suitable for educational and small-scale robotic applications [5].

C. Laboratory Automation Systems

The automation of laboratory processes has gained significant attention in recent years. Automated guided vehicles (AGVs) have been deployed in pharmaceutical laboratories and hospital settings for material transport [6]. However, these systems often involve significant infrastructure costs and complexity. Simpler line following approaches offer a cost-effective alternative for smaller laboratory environments.

There are some specific requirements for material transport robots in laboratory settings, emphasizing the need for reliability, safety features, and ease of path modification. These requirements highlighted the importance of obstacle avoidance capabilities to prevent collisions with laboratory personnel and equipment.

III. SYSTEM DESIGN

A. Overall System Architecture

The proposed line following robot system consists of three main subsystems: the sensing subsystem, the processing and control subsystem, and the actuation subsystem. Fig. 1 illustrates the overall system architecture showing the interconnections between components.

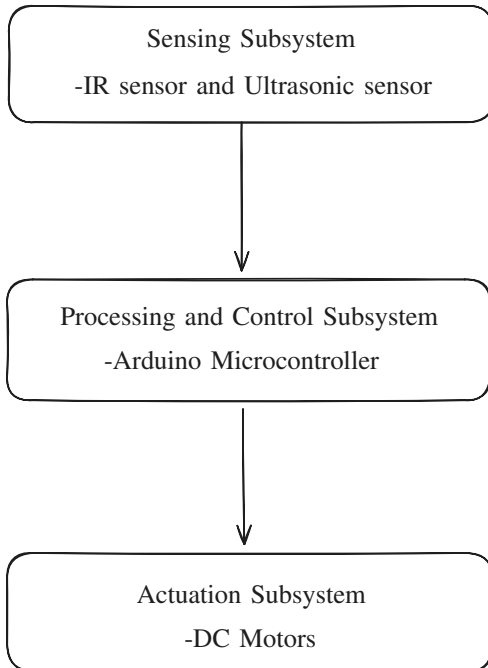


Fig. 1. Overall System Architecture

The sensing subsystem consists of infrared sensors for line detection and ultrasonic sensor for obstacle detection.

The processing subsystem is based on the Arduino Uno microcontroller, which receives sensor inputs, executes control algorithms, and generates appropriate motor control signals. The actuation subsystem includes DC motors with motor driver. These three subsystems work in coordination to achieve autonomous navigation along predefined paths while maintaining safety through obstacle detection and avoidance.

B. Sensing Subsystem

The sensing subsystem is responsible for environmental perception, enabling the robot to detect both the guide path and potential obstacles. This subsystem consists of two complementary sensor types that provide comprehensive situational awareness for safe autonomous operation.

1) *Infrared Sensor Array*: The line detection system utilizes an array of three IR sensor modules arranged in a horizontal configuration at the front of the robot. Each IR sensor module consists of an IR LED emitter and a photodiode receiver to optimize reflectance detection. The sensors are mounted approximately 8 mm above the ground surface to optimize detection performance while maintaining consistent readings across slight variations in floor topology.

The IR sensors operate on the principle of differential reflectivity between the black line and the white surface. When positioned over the white surface, the emitted infrared light reflects back to the receiver with high intensity, producing a high output signal (digital '1'). Conversely, when positioned over the black line, minimal light is reflected due to the absorption characteristics of the black surface, resulting in a low output signal (digital '0'). The digital output from each sensor is read by the Arduino through dedicated digital input pins to determine the robot's position relative to the line [7].



Fig. 2. IR Sensor

The three-sensor configuration provides sufficient resolution for accurate position determination while maintaining system simplicity. The sensors are placed in a way that each sensor spaced approximately 40 mm apart. This spacing allows for detection of standard electrical tape lines while providing adequate overlap for reliable tracking. The centre sensor is aligned with the intended path centreline, while the outer sensors are aligned with surface.

2) *Ultrasonic Sensor*: An HC-SR04 ultrasonic distance sensor is mounted at the front of the robot for obstacle detection and collision avoidance. This sensor operates at 40 kHz ultrasonic frequency and can measure distances from 2 cm to 400 cm with approximately 3 mm resolution [8]. The sensor is positioned at the robot's front face, approximately 90 mm above the ground.



Fig. 3. HC-SR04 Ultrasonic Sensor

When triggered, it emits a burst of eight ultrasonic pulses at 40 kHz. These pulses propagate through air and reflect back when they encounter an obstacle. The sensor measures the time taken for the echo to return. By measuring this time with the Arduino and using the speed of sound, the distance to the obstacle can be calculated.

C. Processing and Control Subsystem

The Arduino Uno microcontroller board serves as the central processing unit and control hub for the entire robotic system. Based on the ATmega328P microcontroller, it provides sufficient computational resources for real-time sensor processing, algorithm execution, and motor control while maintaining low power consumption and cost-effectiveness.

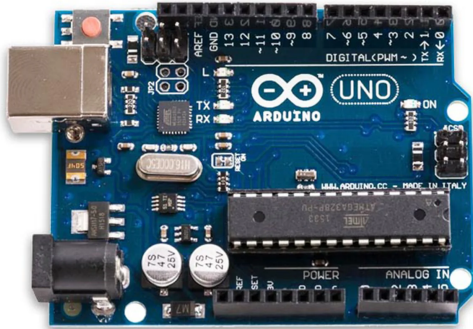


Fig. 4. Arduino UNO Microcontroller

The Arduino Uno features the ATmega328P microcontroller running at 16 MHz clock speed, providing adequate processing power for the control loop execution at approximately 100 Hz. This high update rate ensures responsive control and smooth motion characteristics. The board includes 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog input pins, 32 KB flash memory for program storage, 2 KB SRAM for runtime variables, and 1 KB EEPROM for persistent data storage [9].

D. Actuation Subsystem

The actuation subsystem converts electrical control signals from the Arduino into mechanical motion, enabling the robot to navigate along the designated path. This subsystem consists of the motor driver circuit and DC gear motors configured in a differential drive arrangement.

1) *Motor Driver Module:* The L298N dual H-bridge motor driver module is employed to interface between the low-power Arduino outputs and the high-current DC motors. The H-bridge configuration allows for bidirectional motor control (forward and reverse) with variable speed control through PWM signals. The module can handle continuous currents up to 2A per channel with peak currents of 3A, providing adequate capacity for the selected motors with safety margin.

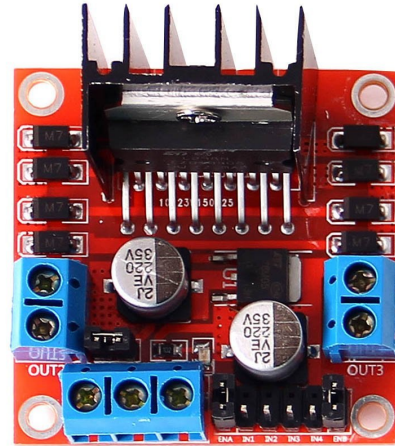


Fig. 5. L298N Motor Driver Module

The motor driver receives four control signals from the Arduino: IN1 and IN2 for direction control of Motor A, IN3 and IN4 for direction control of Motor B, and ENA and ENB for PWM speed control of Motor A and Motor B respectively. By setting appropriate logic levels on the direction pins and varying the PWM duty cycle on the enable pins, the Arduino can independently control each motor's speed and direction. This differential control enables the robot to execute various functions including forward motion, backward motion, left turns, right turns, and pivot turns.

2) *DC Gear Motors:* Two identical DC gear motors with integrated gearboxes provide the mechanical drive power for robot motion. These motors are selected for their appropriate speed-torque characteristics, compact size, compatibility with the available voltage supply and the ability to carry the robot's weight plus payload.

The motors are mounted on opposite sides of the robot chassis in a differential drive configuration. Each motor directly drives one wheel, with the wheels oriented parallel to each other. This configuration, while simple, provides excellent maneuverability including the ability to pivot in place by rotating the motors in opposite directions.

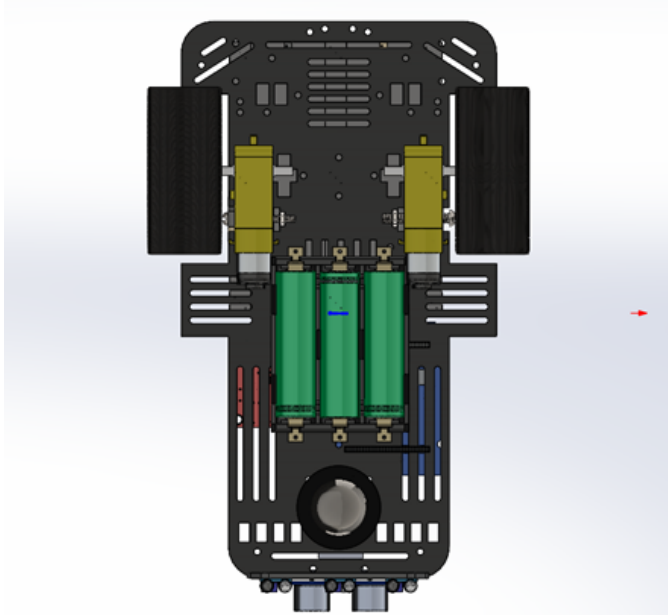


Fig. 6. Bottom View of Chassis

E. Power Supply System

The robot is powered by a 3S Lithium-Ion battery pack (comprising three 18650 cells), providing a nominal 11.1V output to drive the DC motors with high torque and speed. To ensure system stability, the raw battery voltage is sent through a master switch for immediate circuit isolation and then processed by an LM2596 buck converter. This module efficiently steps down the higher input voltage (11.1V – 12.6V) to a regulated 5V supply for the microcontroller and sensor array, effectively decoupling sensitive logic circuits from voltage fluctuations caused by motor load. Mechanically, the 3-cell battery assembly is mounted near to the centroid of the chassis for ensuring balanced traction and preventing tipping during fast turns and movement.

F. Chassis and Mechanical Design

The robot chassis is constructed from Plastic, chosen for its combination of light weight, adequate structural rigidity and low cost. The chassis dimensions are approximately 340 mm x 196 mm x 66 mm (length x width x height), providing sufficient space for all electronic components while maintaining a compact footprint suitable for laboratory corridors and workspaces.

The robot utilizes a single-layer design, featuring one sturdy base plate that holds all mechanical and electronic parts on a single level. The main chassis plate acts as the robot's backbone and features strategic cutouts to reduce weight by approximately 30%. These slots make the robot lighter without sacrificing the structural strength needed for stable movement.

The movement system uses a standard differential drive setup arranged on the main board. Two DC gear motors are securely mounted to the rear of the chassis using rigid brackets, driving high-traction silicone tires (approx. 65 mm diameter). A passive ball caster is placed at the front center,

creating a stable tripod configuration. This three-point contact ensures the robot remains flat on the ground and does not rock during starting or stopping.

To keep the robot balanced on this single board, the 3S Lithium-Ion battery pack (11.1V) is mounted exactly along the middle (centerline) of the chassis. Placing the heavy battery in the center is critical for a flat design; it ensures the Center of Gravity stays low and is shared equally between the drive wheels. This prevents the robot from leaning to one side, which would cause it to drift away from the line.

The Payload Compartment (168 mm x 130 mm x 80 mm) is attached to the main assembly to carry the load. To save space on the single board, the compartment is raised up on 40 mm metal standoffs. This setup effectively creates a "raised deck," allowing the other circuit components and wires to keep protected under the compartment.

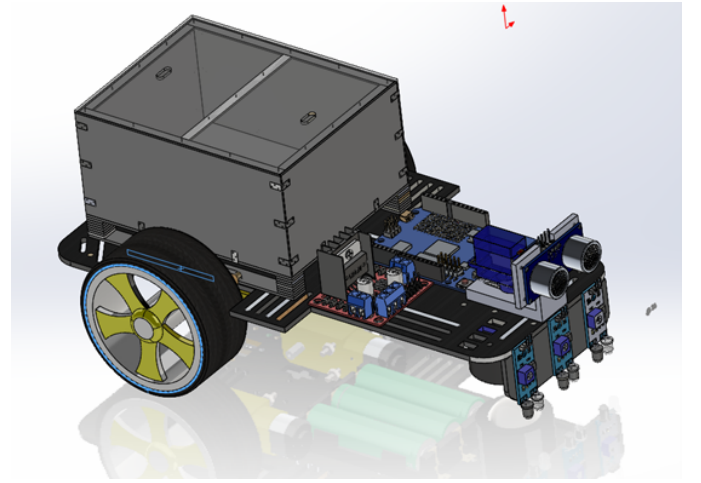


Fig. 7. Top View of Robot

IV. WORKING METHODOLOGY

The control logic was first developed and tested in Python within the Webots environment, allowing rapid prototyping and debugging without hardware constraints. The simulation validated key functionalities including line following accuracy, junction detection reliability, obstacle avoidance response time, and bidirectional navigation capability. Multiple test scenarios were executed to verify system behavior under various conditions such as sharp turns, narrow corridors, and dynamic obstacle placement.

After successful validation in simulation, the same control logic was re-implemented in Arduino C++ to run on the physical robot. This ensured algorithmic consistency while accounting for hardware-specific constraints such as PWM-based motor control and sensor noise characteristics. The simulation-to-hardware transition process confirmed that the control strategies developed in the virtual environment translated effectively to real-world operation, validating the simulation as an effective development tool for autonomous navigation algorithms.

A. System Overview

The robot operates as an autonomous material transport system designed to navigate between a central starting point and three designated laboratory rooms. The system employs a command-based operation model where users specify destination rooms, and the robot autonomously handles navigation, obstacle avoidance, and return journey without further intervention.

The navigation infrastructure consists of black line paths marked on the laboratory floor, forming a junction-based network connecting the starting point to three destination rooms. The robot uses differential line detection and junction recognition to make routing decisions and maintain accurate path tracking.

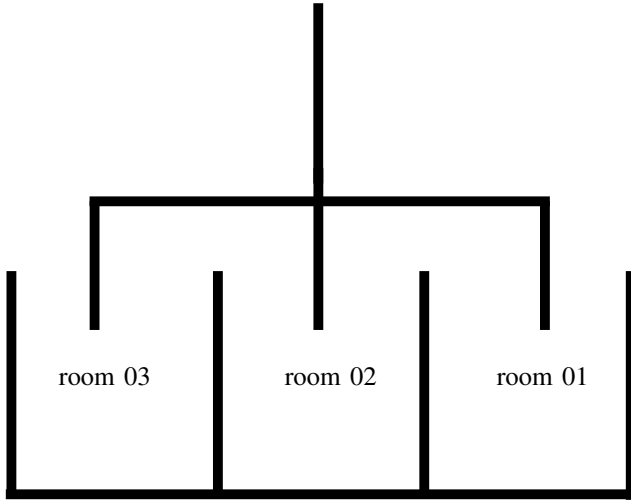


Fig. 8. Floor plan of simulated laboratory

B. Initial State and System Activation

1) *Power ON state*: Upon power-up, the robot enters an IDLE state where all systems are initialized but sensors and motors remain inactive to conserve power and prevent unintended motion. In this state, the Arduino microcontroller completes hardware initialization, establishes serial communication for user interface, and awaits user input commands. This will give enough time for operators to prepare the environment before material transport begins.

2) *Room Selection and Sensor Activation*: When the operator inputs a destination room number (1, 2, or 3) through the user interface (push buttons), the system undergoes following activation sequence.

- Store the Destination: Store the destination room number for future decision makings.
- Sensor Initialization: All three IR sensors are powered on. The ultrasonic sensor is powered on and begins distance monitoring.
- State Transition: The system changes from *IDLE* to line-following states.
- Operation: The robot stores the destination room in memory and begins motion.

C. Line Following Navigation

1) *Three-sensor configuration*: The robot utilizes three IR sensors arranged in a linear array at the front of the chassis. The sensors are designated as Left, Center, and Right, positioned approximately 40 mm apart to match standard electrical tape line width.

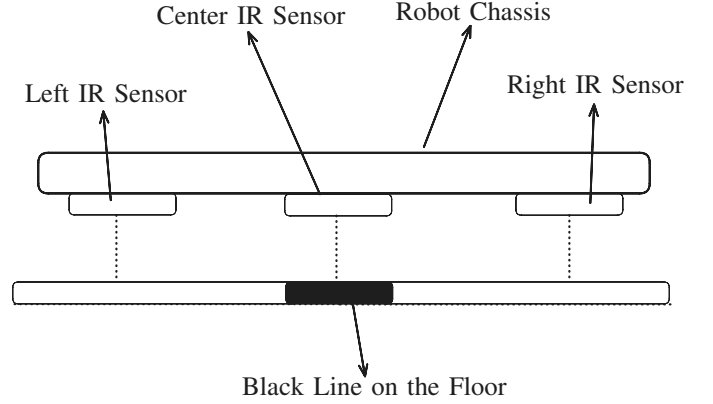


Fig. 9. Sensor Configuration

Each sensor produces a threshold-based logical signal derived from its analog value

Black Line: Sensor value greater than 6 (approximately), Interpreted as line present

White Surface: Sensor value less than or equal to 6 (approximately), Interpreted as no line

2) *Basic Line Following Logic*: The robot implements differential speed control based on the three-sensor readings. The control algorithm continuously evaluates sensor states and adjusts motor speeds to maintain the center sensor aligned with the line centerline. While moving following cases can occur

Left	Center	Right	Position	Action
LOW	LOW	LOW	Line end	Stop
LOW	LOW	HIGH	Left Deviation	Turn Right
HIGH	LOW	LOW	Right Deviation	Turn Left
HIGH	HIGH	HIGH	Junction Detected	Decision
LOW	HIGH	LOW	Centered on Line	Move Forward

HIGH – Black line detected
LOW – White surface detected

```
// Line following logic
else if ((left_ir_value > right_ir_value) && (left_ir_value > 6 && left_ir_value < 15)) {
    Serial.println("Go left");
    left_speed = -max_speed * 0.4;
    right_speed = right_speed; // Keep right at default
}
else if ((right_ir_value > left_ir_value) && (right_ir_value > 6 && right_ir_value < 15)) {
    Serial.println("Go right");
    left_speed = left_speed; // Keep left at default
    right_speed = -max_speed * 0.4;
}
```

Fig. 10. Code for line following

D. Junction Detection and Room Navigation

1) *Junction Recognition*: A critical feature of the multi-room navigation system is the ability to detect and respond

to path junctions. Junctions are identified when all three IR sensors simultaneously detect the black line (for our case), indicating the robot has encountered an intersection where multiple paths converge.

```
// Junction detection - all three sensors detect line
if ((centre_ir_value > 10 && centre_ir_value < 50) &&
    (left_ir_value > 10 && left_ir_value < 50) &&
    (right_ir_value > 10 && right_ir_value < 50)) {

    Serial.println("-----");
    junction_count++;
}
```

Fig. 11. Condition to detect the junction

This three-sensor simultaneous activation provides reliable junction identification, distinguishing junctions from regular line following where typically only one or two sensors detect the line.

2) *Room-Specific Navigation Strategy*: Upon junction detection, the robot executes a predetermined turning maneuver based on the destination room which is given as input by the operator. The navigation strategy employs differential drive pivot turns to change direction while minimizing positional errors.

Room 1 – Make left turn

Room 2 – Go straight

Room 3 – Make right turn

3) *Post Junction Operation*: After completing the turning maneuver according to the destination room, the robot resumes normal line following mode, using the three-sensor array to track the branch path leading to the destination room. The control system automatically transitions back to the line following algorithm described above.

E. Continuous Obstacle Avoidance

Throughout the entire journey (forward and return paths), the ultrasonic distance sensor operates continuously in parallel with the line following system, providing real-time obstacle detection and collision prevention.

```
float getDistance() {
    // Send ultrasonic pulse
    digitalWrite(ultrasonic_trig_pin, LOW);
    delayMicroseconds(2);
    digitalWrite(ultrasonic_trig_pin, HIGH);
    delayMicroseconds(10);
    digitalWrite(ultrasonic_trig_pin, LOW);

    // Read echo pulse duration
    long duration = pulseIn(ultrasonic_echo_pin, HIGH);

    // Calculate distance in cm (speed of sound = 343 m/s)
    float distance = duration * 0.034 / 2;

    return distance;
}
```

Fig. 12. Code to calculate the distance for obstacle

The obstacle detection algorithm runs as a high-priority concurrent task that can interrupt line following at any moment when a potential collision is detected. This architecture ensures

safety takes precedence over navigation objectives. The system employs a safety threshold distance of 20 cm, providing adequate reaction time for the robot to stop before collision at typical operating speeds.

```
// Obstacle detected (distance < 20cm)
if (distance < SAFETY_DISTANCE) {

    // First detection
    if (!blocked) {
        Serial.println("\n!!! OBSTACLE DETECTED - EMERGENCY STOP !!!");
        Serial.print("Distance: ");
        Serial.print(distance);
        Serial.println(" cm\n");

        blocked = true;
        block_start = now;
        last_check = now;
        alert_on = false;
    }
}
```

Fig. 13. Condition to check if the obstacle within 20 cm

When an obstacle is detected within the 20 cm safety zone, the robot executes an immediate emergency stop and enters a waiting mode until the obstacle is cleared. Then the system checks for obstacle for every 500 ms while in the stopped condition. If obstacle is removed, then it will resume the current operation.

If an obstacle persists for more than 30 seconds, the system generates an alert (buzzer) and waits for manual intervention, preventing indefinite waiting in case of permanent obstructions.

```
// Timeout after 30 seconds
if (elapsed >= 30.0 && !alert_on) {
    Serial.println("\n*** ALERT: Obstacle >30s - Manual intervention needed ***\n");
    alert_on = true;
    digitalWrite(buzzer_pin, HIGH); // Turn on buzzer
}
}
```

Fig. 14. Code to determine if manual intervention is needed

F. Destination Arrival and Payload Loading

1) *End of Line Detection*: The robot determines it has reached the destination room by detecting the termination of the black line path. This condition is identified when all three IR sensors simultaneously read LOW for a sustained period, indicating the robot has traveled beyond the line endpoint.

2) *Loading State*: Upon arrival at the destination room, the robot executes the following sequence:

1. Both motors set to zero velocity
2. Maintain current position
3. System waits for worker input

While the robot is stationary in loading state, laboratory personnel manually place materials on the robot's transport platform. The platform design accommodates standard laboratory items with maximum payload capacity of 1.2 kg.

G. Return Journey Initiation

After payload loading is complete, the laboratory worker initiates the return journey by pressing a designated Return button. This manual confirmation ensures the robot only begins

the return trip when the operator is ready and has cleared the path. To navigate back along the same path, the robot must reverse its orientation by executing a 180° pivot turn in place. This maneuver is implemented as a timed rotation using differential drive control.

H. Return Path Navigation

1) *Reverse Path Following*: The return journey employs the same line following algorithm as the forward path, with the robot now traveling in the opposite direction along the same line network. The three-sensor line following logic operates identically.

2) *Junction Handling on Return Path*: A critical difference in return navigation is the junction turning behavior. Since the robot is traveling in the reverse direction, the turning commands at junctions must be inverted to follow the correct path back to the starting point.

Room	Junction Decision in Forward Path	Junction Decision in Reverse Path
Room 1	Left turn	Right turn
Room 2	Go straight	Go straight
Room 3	Right turn	Left turn

This path reversal ensures the robot navigates back through the junction network to return to the starting point, completing the bidirectional transport cycle.

3) *Return Destination Detection*: The robot recognizes arrival at the starting point using the same end-of-line detection method as the forward journey. When all three sensors read LOW for a sustained period, indicating the line has terminated, the system concludes the return journey is complete. Then the robot will make a 180° turn and wait for the next input.

V. CONCLUSION

A. Significance of the Project

The line following robot for laboratory material transport represents a significant advancement in addressing the challenges of manual material handling in educational and research facilities. Laboratory environments often require frequent transportation of materials, equipment, and samples between multiple locations, which is time-consuming and diverts skilled personnel from their core research activities. Our robot addresses these limitations by providing an autonomous, cost-effective solution that operates efficiently in structured indoor environments where precise material delivery is essential.

One of the major challenges in laboratory automation is the high cost of commercial automated guided vehicles (AGVs) and require complex infrastructure. To combat this, we designed a low-cost system using readily available components including Arduino microcontroller, infrared sensors, and standard DC motors. This affordability makes advanced automation accessible to smaller educational institutions and research laboratories with limited budgets.

Furthermore, while most laboratory transport solutions require extensive infrastructure modifications or operate with limited autonomy, our robot's fully autonomous line following

capability with multi-room navigation represents a practical advancement. By utilizing simple black line markings on floors and incorporating junction-based routing with continuous obstacle avoidance, the system enhances both operational efficiency and safety. The robot can navigate to three different laboratory rooms, wait for payload loading, and autonomously return to the starting point without human intervention.

B. Suggestions for Future Research

Future research can advance this system through several promising directions. Technologies like computer vision and SLAM (Simultaneous Localization and Mapping) can enhance the robot's ability to navigate environments without relying solely on line markings, enabling operation in unmarked areas and dynamic path planning. The system could also benefit from improved power management strategies including higher capacity lithium-ion batteries and energy-efficient motor control algorithms to extend operational time for continuous laboratory operations.

Furthermore, integrating the robot with advanced user interfaces such as mobile applications and voice control systems can significantly improve usability and accessibility for laboratory personnel. Database integration with automatic transport logging would enable real-time monitoring, historical analysis, and seamless integration with Laboratory Information Management Systems for comprehensive sample tracking and regulatory compliance. These features would transform the robot from a standalone transport device into an intelligent component of a smart laboratory ecosystem.

REFERENCES

- [1] I. Hassanzadeh and S. Akbari, "Application of autonomous mobile robots in health care systems," in *Proc. 2nd International Conference on Control, Instrumentation and Automation*, 2011, pp. 628–633.
- [2] M. Pakdaman and M. M. Sanaatiyan, "Design and Implementation of Line Follower Robot," *2009 Second International Conference on Computer and Electrical Engineering*, Dubai, United Arab Emirates, 2009, pp. 585–590, doi: 10.1109/ICCEE.2009.43.
- [3] R. C. Luo and C.-C. Chang, "Multisensor Fusion and Integration: A Review on Approaches and Its Applications in Mechatronics," in *IEEE Transactions on Industrial Informatics*, vol. 8, no. 1, pp. 49–60, Feb. 2012, doi: 10.1109/TII.2011.2173942.
- [4] S. Kamalasadan and A. Hande, "A PID Controller for Real-Time DC Motor Speed Control using the C505C Microcontroller," in *CAINE*, 2004, pp. 34–39.
- [5] T. Mahmud et al., "Design and Implementation of an Ultrasonic Sensor-Based Obstacle Avoidance System for Arduino Robots," *2023 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD)*, Dhaka, Bangladesh, 2023, pp. 264–268, doi: 10.1109/ICICT4SD59951.2023.10303550.
- [6] S. Benzidia, B. Ageron, O. Bentahar, and J. Husson, "Investigating automation and AGV in healthcare logistics: a case study based approach," *International Journal of Logistics Research and Applications*, vol. 22, no. 3, pp. 273–293, 2019.
- [7] S. S. Suryawan, S. M. Musamwar, S. R. Kolhe, S. A. Tengane, S. S. Hanumate, and P. H. Sahare, "Line follower and obstacles avoider robot," *International Research Journal of Engineering and Technology (IRJET)*, pp. 460–468, 2019.
- [8] V. A. Zhmud, N. O. Kondratiev, K. A. Kuznetsov, V. G. Trubin, and L. V. Dimitrov, "Application of ultrasonic sensor for measuring distances in robotics," *Journal of Physics: Conference Series*, vol. 1015, p. 032189, May 2018, doi: 10.1088/1742-6596/1015/3/032189.
- [9] J. M. Hughes, *Arduino: a technical reference: a handbook for technicians, engineers, and makers*. O'Reilly Media, Inc., 2016.